# NLP Answer Sheet

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**Q1. What is POS Tagging? Why is it important?**

POS (Part-of-Speech) tagging is a core task in Natural Language Processing (NLP) that involves assigning grammatical categories to each word in a sentence. These categories include noun, verb, adjective, pronoun, adverb, etc. POS tagging helps machines understand sentence structure and meaning.

**Key Concepts in POS Tagging:**
• Parts of Speech: Categories like nouns, verbs, adjectives, adverbs, pronouns, etc. that define the role of a word in a sentence.
• Tagging: The process of assigning a specific part-of-speech label to each word in a sentence.
• Corpus: A large collection of text data used to train POS taggers.

**Types of POS Tagging:**
1. Rule-Based POS Tagging: Uses handcrafted linguistic rules to assign tags.
2. Stochastic POS Tagging: Uses probabilistic models such as Hidden Markov Models (HMM) to predict tags.
3. Transformation-Based POS Tagging: Combines rule-based and statistical approaches.
4. Neural POS Tagging: Uses deep learning and neural networks for tagging.

**Why is POS Tagging Important?**
• Syntactic Analysis: POS tags help in understanding the grammatical structure of a sentence, which is essential for parsing.

**Example:**
Sentence: "The cat sat on the mat."
POS Tags: The/DT cat/NN sat/VBD on/IN the/DT mat/NN
Analysis: The tags show that "cat" is a noun, "sat" is a verb, and "on" is a preposition.

• Semantic Understanding: Knowing the role of a word in a sentence helps in understanding meaning.

**Ambiguity Example:**
Sentence: "I saw the man with the telescope."
POS Tags: I/PRP saw/VBD the/DT man/NN with/IN the/DT telescope/NN
Analysis: POS tags help clarify whether "with the telescope" modifies "saw" or "the man".

• Disambiguation: Many words in natural language are ambiguous and can belong to multiple parts of speech depending on context.